

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO3

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 3

Unit Title: Hierarchical and Density-Based Clustering

Unit Mapped RBT: L3 (Apply)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Compare agglomerative clustering with divisive clustering. (10 Marks)

- (a) State the direction in which each of the two builds, and where each one begins.
 - (b) Use the cost of each to state which of the two is used in practice, and give the reason.
- [L3 - Apply]

Q2. Apply Ward linkage to a hierarchical clustering and describe the result it gives. (10 Marks)

- (a) State what Ward linkage minimises at each merge.
 - (b) Use this to state the shape of cluster it produces, and when you would prefer it to single linkage.
- [L3 - Apply]

Q3. Examine a dendrogram on which almost every merge takes place at a similar height. (10 Marks)

- (a) State what the height of a merge represents on the diagram.
 - (b) Use the diagram to decide what you would report about the structure of this dataset.
- [L3 - Apply]

Q4. Use the k-distance plot to choose a value of eps for DBSCAN. (10 Marks)

- (a) Describe how the plot is built from the data.
 - (b) State where the value of eps is read from, and what you would check after running DBSCAN with it.
- [L3 - Apply]

Q5. Apply DBSCAN to a dataset that holds one dense cluster together with one much sparser cluster. (10 Marks)

- (a) State how a point is decided to be a core point, a border point or noise.
 - (b) Use this to state what happens to the sparser cluster, and what could be done about it.
- [L3 - Apply]

Q6. Compare k-Means, agglomerative clustering and DBSCAN on the input each one needs before it can run. (10 Marks)

- (a) State for each of the three whether the number of clusters must be supplied, and what other parameters are required.
 - (b) Use this comparison to select a method for a dataset of unknown shape containing many stray records.
- [L3 - Apply]

Q7. Demonstrate how a border point and a noise point are told apart in DBSCAN. (10 Marks)

- (a) State the definition of a core point, a border point and a noise point.
 - (b) Use a worked example with MinPts set to 4 to show how one particular point is classified.
- [L3 - Apply]

Q8. Relate the choice of linkage criterion to the presence of stray records in the data. (10 Marks)

- (a) State how single linkage and how complete linkage each react to one stray record.
 - (b) Use this to recommend a linkage criterion for a dataset known to contain measurement errors.
- [L3 - Apply]

Q9. Apply agglomerative hierarchical clustering to a small matrix of pairwise distances. (10 Marks)

- (a) State the steps that are followed, and what is recorded at each merge.
- (b) Use the completed dendrogram to obtain a partition into three clusters.

[L3 - Apply]

Q10. Examine the effect of changing MinPts while the value of eps is held fixed. (10 Marks)

- (a) State what MinPts controls in the definition of a core point.
- (b) Use this to describe what happens to the clusters and to the count of noise points as MinPts is raised.

[L3 - Apply]

----- End of Sample Question Set -----